

# Locally Thermal from Globally Athermal: Correlations as a Thermodynamic Resource under Gibbs-Preserving Operations

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## Abstract

We study bipartite quantum states whose marginals are fixed to local Gibbs states. This locally thermal (LT) setting removes all local athermality, so any deviation from the product Gibbs state is stored entirely in correlations. The report has three objectives: to describe the geometry of LT states, to identify the thermodynamic meaning of correlations on LT, and to compare global and local Gibbs-preserving convertibility.

The main analytic target is the full two-qubit LT body. We show that LT is an affine slice of the positive semidefinite cone, hence a spectrahedron, and derive the exact dimension formula  $\dim_{\mathbb{R}} \text{LT}(\gamma_A, \gamma_B) = (d_A^2 - 1)(d_B^2 - 1)$ . For symmetric qubits this yields a 9-dimensional problem. We obtain an explicit normal form for all two-qubit LT states and reduce generic positivity to a Schur-complement criterion. Supporting analytic warmups include diagonal- $T$  slices and LT rays, which provide exact feasibility bounds and clean numerical probes. In dimension  $d = 3$ , where the full LT body is 64-dimensional, we treat the commuting qutrit LT subclass exactly as a transportation polytope with mutual-information range  $0 \leq I \leq H(g)$ .

Operationally, we prove that on LT the free-energy monotone relative to the product Gibbs state coincides with mutual information, and that LT is closed under local Gibbs-preserving maps. We also exhibit the thermofield-double to dephased family as an explicit ordered chain under local GP. Finally, we formulate the membership, projection, and convertibility problems as semidefinite programs and explain why local convertibility requires relaxations or verification because of the nonconvex product-channel constraint.

# 1 Introduction

This project studies bipartite quantum states whose local marginals are already thermal. For a symmetric system  $AA'$  with local Hamiltonian  $H$  and Gibbs state

$$\gamma = \frac{e^{-\beta H}}{Z}, \quad \Gamma := \gamma \otimes \gamma,$$

a state  $\rho_{AA'}$  is *locally thermal* (LT) when

$$\mathrm{Tr}_{A'} \rho_{AA'} = \gamma, \quad \mathrm{Tr}_A \rho_{AA'} = \gamma.$$

Local athermality is then absent by construction, so any nonequilibrium resource relative to  $\Gamma$  is necessarily stored in correlations.

This makes LT a useful constrained setting for quantum thermodynamics. In the unrestricted bipartite theory, free energy, local nonequilibrium, coherence, and correlations can all vary at once. On LT, by contrast, the local energetic content is frozen to equilibrium and the problem becomes: what global structures remain, how large is the allowed state space, and what state transformations are still possible under Gibbs-preserving dynamics. The report addresses three linked questions:

1. What is the geometry of the LT set?
2. How does correlation resource behave under global Gibbs-preserving (GP) operations?
3. What changes when one restricts to *local* GP maps?

The project is organised around one main target and two supporting strands.

**Main target: full two-qubit LT characterisation.** For  $d = 2$ , the LT set has real dimension  $(2^2 - 1)^2 = 9$ . This is the first case where the full constrained state space is nontrivial but still analytically manageable. The main structural result is therefore an explicit normal form for all two-qubit LT states together with a Schur-complement reduction of the positivity problem.

**Supporting strand 1: warmup slices and rays.** Before analysing the full 9-dimensional body, it is useful to study tractable subfamilies: diagonal- $T$  states, commuting states, and one-parameter LT rays. These give exact feasibility tests, interpretable geometry, and numerically stable convertibility probes. They are not the main theorem, but they explain how the full problem was approached.

**Supporting strand 2: higher-dimensional structured subclasses.** For qutrits, the full LT set already has dimension  $(3^2 - 1)^2 = 64$ , so a comparably explicit classification is unrealistic. The meaningful higher-dimensional contribution is instead the commuting qutrit LT subclass, where the problem reduces to a transportation polytope with exact correlation bounds.

The three capstone objectives are therefore:

1. **Structure:** describe LT as a constrained convex set and characterise the low-dimensional cases as explicitly as possible.
2. **Resource meaning:** show that on LT the thermodynamic resource relative to  $\Gamma$  is exactly correlation.
3. **Convertibility:** compare global GP and local GP state transformations analytically and numerically.

The main results supporting these objectives are:

1. LT is an affine slice of the positive semidefinite cone, hence a spectrahedron.
2.  $\dim_{\mathbb{R}} \text{LT}(\gamma_A, \gamma_B) = (d_A^2 - 1)(d_B^2 - 1)$ .
3. Every two-qubit LT state admits an explicit 9-parameter matrix form.
4. Generic two-qubit LT positivity reduces to explicit Schur-complement inequalities.
5. On LT,

$$D(\rho \| \Gamma) = I(A : A')_{\rho}.$$

6. LT is closed under local GP maps.
7. The diagonal- $T$  warmup family admits exact positivity and separability criteria.
8. The commuting qutrit LT subclass is a 4-dimensional transportation polytope with exact range  $0 \leq I \leq H(g)$ .
9. The TFD-to-dephased family gives an explicit ordered chain under local GP.

## 2 Core framework

### 2.1 Locally thermal states as a constrained convex body

For fixed local Gibbs states  $\gamma_A$  and  $\gamma_B$ , define

$$\text{LT}(\gamma_A, \gamma_B) = \{\rho_{AB} \in \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B) : \text{Tr}_B \rho_{AB} = \gamma_A, \text{Tr}_A \rho_{AB} = \gamma_B\}.$$

Writing  $\Gamma := \gamma_A \otimes \gamma_B$ , every LT state has the form

$$\rho_{AB} = \Gamma + C,$$

where  $C = C^\dagger$  satisfies

$$\text{Tr}_A C = 0, \quad \text{Tr}_B C = 0,$$

and positivity becomes the single nonlinear constraint  $\Gamma + C \succeq 0$ . Hence

$$\text{LT}(\gamma_A, \gamma_B) = (\Gamma + \mathcal{Z}) \cap \text{PSD},$$

with

$$\mathcal{Z} := \{C = C^\dagger : \text{Tr}_A C = 0, \text{Tr}_B C = 0\}.$$

This immediately shows that LT is a *spectrahedron*: an affine slice of the positive semidefinite cone. In the convex-algebraic-geometric language used for semidefinite programming, a spectrahedron is exactly a feasible set cut out by a linear matrix inequality; see, for example, Helton and Vinnikov [17].

**Proposition 2.1** (Dimension of the LT set).

$$\dim_{\mathbb{R}} \text{LT}(\gamma_A, \gamma_B) = (d_A^2 - 1)(d_B^2 - 1).$$

*In particular,  $\dim_{\mathbb{R}} \text{LT} = 9$  for qubits and 64 for qutrits.*

*Proof.* A density matrix on  $\mathcal{H}_A \otimes \mathcal{H}_B$  has  $(d_A d_B)^2 - 1$  real parameters. Fixing the two marginals imposes  $(d_A^2 - 1) + (d_B^2 - 1)$  independent real linear constraints, so

$$(d_A d_B)^2 - 1 - (d_A^2 - 1) - (d_B^2 - 1) = (d_A^2 - 1)(d_B^2 - 1).$$

□

### 2.2 Geometric language used later

Two pieces of convex-geometric language appear repeatedly in the report.

A *polytope* is a bounded convex set that can be written equivalently either as the convex hull of finitely many points or as a bounded intersection of finitely many affine half-spaces. The commuting qutrit LT family will be a polytope because it is described only by finitely many linear equalities and inequalities on classical probabilities.

A *spectrahedron* is the semidefinite analogue: it is a set of the form

$$\{x \in \mathbb{R}^n : A_0 + x_1 A_1 + \cdots + x_n A_n \succeq 0\},$$

with fixed Hermitian matrices  $A_i$ . LT is of this type because the marginal constraints are linear and positivity is a PSD condition. This distinction matters later: polytopes have flat faces coming from linear inequalities, whereas spectrahedra can have genuinely curved semidefinite boundaries.

## 2.3 Correlation-tensor parameterisation and provenance of the qubit formulas

Let  $\{F_\mu\}_{\mu=1}^{d_A^2-1}$  and  $\{G_\nu\}_{\nu=1}^{d_B^2-1}$  be orthonormal traceless Hermitian operator bases. Every zero-marginal Hermitian operator  $C \in \mathcal{Z}$  can then be expanded as

$$C = \sum_{\mu=1}^{d_A^2-1} \sum_{\nu=1}^{d_B^2-1} c_{\mu\nu} F_\mu \otimes G_\nu, \quad c_{\mu\nu} \in \mathbb{R}.$$

For qubits this becomes a real  $3 \times 3$  correlation tensor  $T$ ; for qutrits, a real  $8 \times 8$  tensor. This is where the  $(d_A^2 - 1)(d_B^2 - 1)$  parameter count comes from.

For two qubits, the standard external starting point is the Bloch/Fano decomposition

$$\rho = \frac{1}{4} \left( I \otimes I + \vec{r} \cdot \vec{\sigma} \otimes I + I \otimes \vec{s} \cdot \vec{\sigma} + \sum_{i,j \in \{x,y,z\}} T_{ij} \sigma_i \otimes \sigma_j \right),$$

used throughout the two-qubit literature; one early standard reference is Horodecki, Horodecki, and Horodecki [9]. The LT constraint does not introduce a different representation; it simply fixes the local Bloch vectors. For symmetric thermal marginals

$$\gamma = \begin{pmatrix} g & 0 \\ 0 & 1 - g \end{pmatrix}, \quad z_\gamma := 2g - 1,$$

one has

$$\vec{r} = \vec{s} = (0, 0, z_\gamma).$$

Substituting this into the standard Bloch/Fano form gives the report's LT-specific Pauli

parameterisation

$$\rho = \frac{1}{4} \left( I \otimes I + z_\gamma (\sigma_z \otimes I + I \otimes \sigma_z) + \sum_{i,j \in \{x,y,z\}} T_{ij} \sigma_i \otimes \sigma_j \right).$$

So this equation is not an isolated formula pulled from nowhere: its source is the standard two-qubit Bloch/Fano representation, and the LT conditions simply fix the one-body part. Restricting further to diagonal  $T = \text{diag}(t_x, t_y, t_z)$  gives an  $X$ -state. The class of two-qubit  $X$ -states is standard in the literature; see Yu and Eberly [10] and Ali, Rau, and Alber [11]. Again, the exact matrix written later in the report is the LT-specialised version of that standard  $X$ -state idea: diagonal  $T$  kills all matrix entries except the diagonal and anti-diagonal ones.

## 2.4 Thermodynamic structure on LT

The project uses two standard information-theoretic quantities:

$$S(\rho) := -\text{Tr}(\rho \log \rho), \quad D(\rho \parallel \sigma) := \text{Tr}[\rho(\log \rho - \log \sigma)],$$

and mutual information

$$I(A:B)_\rho = S(\rho_A) + S(\rho_B) - S(\rho_{AB}) = D(\rho_{AB} \parallel \rho_A \otimes \rho_B).$$

For a Gibbs state  $\gamma = e^{-\beta H}/Z$ , relative entropy to equilibrium satisfies

$$D(\rho \parallel \gamma) = \beta(F(\rho) - F(\gamma)).$$

Thus  $D(\rho \parallel \gamma)$  measures nonequilibrium free-energy content.

**Proposition 2.2** (Athermality equals correlation on LT). *If  $\rho \in \text{LT}(\gamma_A, \gamma_B)$ , then*

$$D(\rho \parallel \gamma_A \otimes \gamma_B) = I(A:B)_\rho.$$

*In the symmetric case,  $D(\rho \parallel \Gamma) = I(A:A')_\rho$ .*

*Proof.* Use the identity

$$I(A:B)_\rho = D(\rho \parallel \rho_A \otimes \rho_B)$$

and substitute  $\rho_A = \gamma_A$  and  $\rho_B = \gamma_B$ . □

This identity is central to the whole report: once the local marginals are fixed to thermal equilibrium, the global free-energy excess above  $\Gamma$  is exactly the total correlation.

## 2.5 How mutual information is evaluated analytically and numerically

In the analytic derivations, the mutual information is always the von Neumann quantity

$$I(A:B)_\rho = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

On LT this simplifies because the marginals are fixed:

$$I(A:B)_\rho = S(\gamma_A) + S(\gamma_B) - S(\rho_{AB}).$$

For symmetric qubits with  $\gamma = \text{diag}(g, 1 - g)$  this becomes

$$I(A:A')_\rho = 2H_2(g) - S(\rho),$$

where  $H_2(g) = -g \log g - (1 - g) \log(1 - g)$  is the binary entropy.

Numerically, the code evaluates  $S(\rho)$  from the eigenvalues  $\{\lambda_k\}$  of the density matrix:

$$S(\rho) = - \sum_k \lambda_k \log \lambda_k.$$

The same eigenvalue formula is used for the marginals. In the commuting qutrit subclass, the state is classical from the start, so the von Neumann entropy reduces to Shannon entropy and mutual information becomes

$$I(P) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{g_i g_j} = 2H(g) - H(P).$$

This is the exact equation implemented whenever the state is represented by its probability matrix  $P$  rather than by a full density matrix.

## 2.6 Gibbs-preserving maps and local closure

A channel  $\Phi$  is *Gibbs-preserving* (GP) when  $\Phi(\gamma) = \gamma$ . By data processing,

$$D(\rho||\gamma) \geq D(\Phi(\rho)||\Phi(\gamma)),$$

so every GP map makes  $D(\rho||\gamma)$  monotone.

For a bipartite system, *local GP* means a tensor-product channel  $\Phi_A \otimes \Phi_B$  with

$$\Phi_A(\gamma_A) = \gamma_A, \quad \Phi_B(\gamma_B) = \gamma_B.$$

**Proposition 2.3** (Closure of LT under local GP). *If  $\rho \in \text{LT}(\gamma_A, \gamma_B)$  and  $\Phi_A, \Phi_B$  are local GP channels, then*

$$(\Phi_A \otimes \Phi_B)(\rho) \in \text{LT}(\gamma_A, \gamma_B).$$

*Proof.* Taking partial traces,

$$\text{Tr}_B[(\Phi_A \otimes \Phi_B)(\rho)] = \Phi_A(\text{Tr}_B \rho) = \Phi_A(\gamma_A) = \gamma_A,$$

and similarly for the  $B$  marginal. □

This proposition is the basic reason local GP convertibility is meaningful on LT: the allowed dynamics do not leave the LT manifold.

### 3 Main analytic results

#### 3.1 Full two-qubit LT characterisation

For symmetric qubits let

$$\gamma = \begin{pmatrix} g & 0 \\ 0 & 1-g \end{pmatrix}, \quad \frac{1}{2} \leq g < 1, \quad \Gamma = \gamma \otimes \gamma.$$

The qubit case is the main analytic target because LT has dimension 9: large enough to be genuinely structural, but still small enough to analyse directly.

**Proposition 3.1** (Canonical two-qubit LT form). *A two-qubit state is LT with both marginals equal to  $\gamma$  if and only if in the product energy basis*

$$\rho = \begin{pmatrix} g-a & x & y & z \\ x^* & a & w & -y \\ y^* & w^* & a & -x \\ z^* & -y^* & -x^* & 1-g-a \end{pmatrix}, \quad a \in \mathbb{R}, \quad x, y, z, w \in \mathbb{C}, \quad (1)$$

with  $\rho \succeq 0$ .

*Proof.* Start from a general  $4 \times 4$  Hermitian density matrix and impose the linear marginal constraints  $\text{Tr}_A \rho = \gamma$  and  $\text{Tr}_B \rho = \gamma$ . This leaves  $15 - 6 = 9$  real degrees of freedom and yields (1).  $\square$

#### 3.2 Immediate bounds and local gauge fixing

Positivity of the diagonal entries and the simplest principal minors already gives useful necessary bounds:

$$0 \leq a \leq \min(g, 1-g), \quad |x|^2 \leq a(g-a), \quad |y|^2 \leq a(g-a),$$

$$|w|^2 \leq a(1-g-a), \quad |z|^2 \leq (g-a)(1-g-a).$$

These are not sufficient, but they identify the natural scale of the LT parameters and explain why the physically allowed region is already strongly constrained before the full positivity condition is imposed.

The local unitaries that preserve a nondegenerate qubit Gibbs state are diagonal phase rotations in the energy basis. They preserve both LT and positivity, and they act on the

free parameters by rephasing  $x$  and  $y$ . Hence one may choose a convenient gauge in which

$$x, y \in \mathbb{R}_{\geq 0}.$$

This does not reduce the dimension of LT itself; it only uses local GP symmetry to simplify the positivity analysis.

Setting

$$\alpha := g - a, \quad \delta := 1 - g - a,$$

write

$$\rho = \begin{pmatrix} A & B \\ B^\dagger & D \end{pmatrix}, \quad A = \begin{pmatrix} \alpha & x \\ x & a \end{pmatrix}, \quad B = \begin{pmatrix} y & z \\ w & -y \end{pmatrix}, \quad D = \begin{pmatrix} a & -x \\ -x & \delta \end{pmatrix}.$$

Let  $\Delta := \det A = a\alpha - x^2$ .

**Theorem 3.2** (Generic positivity criterion). *Assume  $\Delta > 0$ . Then  $\rho \succeq 0$  if and only if*

$$0 \leq a \leq 1 - g, \quad \Delta \geq 0, \quad N := D - B^\dagger A^{-1} B \succeq 0,$$

where

$$A^{-1} = \frac{1}{\Delta} \begin{pmatrix} a & -x \\ -x & \alpha \end{pmatrix}.$$

Since  $N$  is  $2 \times 2$  Hermitian, this is equivalent to

$$N_{11} \geq 0, \quad N_{22} \geq 0, \quad N_{11}N_{22} \geq |N_{12}|^2.$$

Moreover the entries are explicitly

$$N_{11} = a - \frac{ay^2 + \alpha|w|^2 - 2xy \Re(w)}{\Delta}, \tag{2}$$

$$N_{22} = \delta - \frac{\alpha y^2 + a|z|^2 + 2xy \Re(z)}{\Delta}, \tag{3}$$

$$N_{12} = \frac{-x\Delta - ayz + \alpha yw + xzw - xy^2}{\Delta}. \tag{4}$$

*Proof.* For  $\Delta > 0$ , the block  $A$  is invertible. The Schur-complement criterion gives

$$\rho \succeq 0 \iff A \succeq 0 \text{ and } D - B^\dagger A^{-1} B \succeq 0.$$

For a  $2 \times 2$  Hermitian matrix, positivity is equivalent to nonnegative diagonal entries and nonnegative determinant.  $\square$

This is the main structural theorem of the report. It turns the full two-qubit LT positivity

problem into explicit inequalities rather than an abstract positivity condition on a  $4 \times 4$  matrix.

### 3.3 Boundary case $\Delta = 0$

The rank-drop boundary  $\Delta = 0$  is not a separate physical model; it is the boundary of the principal block  $A$ . On that boundary one replaces the inverse by the Moore–Penrose pseudoinverse. The appropriate criterion is

$$\rho \succeq 0 \iff A \succeq 0, \quad (I - AA^+)B = 0, \quad D - B^\dagger A^+ B \succeq 0.$$

Equivalently, positivity on the boundary is the continuous closure of the generic inequalities as  $\Delta \downarrow 0$ , together with the range condition  $\text{Ran}(B) \subseteq \text{Ran}(A)$ . The report does not need the full rank-one boundary algebra in the main body, but it is useful to state the correct criterion because it explains what happens when the principal block loses invertibility.

### 3.4 Why slices and rays were used first

The project started with slices and rays because they isolate analytically controlled parts of the LT body before attacking the full 9-parameter problem.

**Classical diagonal qubit LT line.** Restricting to commuting qubit LT states gives

$$\rho_{\text{cl}} = \text{diag}(q, g - q, g - q, 1 - 2g + q),$$

with

$$\max(0, 2g - 1) \leq q \leq g.$$

This is the simplest LT subfamily: it already shows how fixed marginals impose a transportation-type constraint, but it contains no quantum coherence.

**Diagonal- $T$  warmup.** Every two-qubit LT state can be written in Pauli form

$$\rho = \frac{1}{4} \left( I \otimes I + z_\gamma (\sigma_z \otimes I + I \otimes \sigma_z) + \sum_{i,j \in \{x,y,z\}} T_{ij} \sigma_i \otimes \sigma_j \right), \quad z_\gamma = 2g - 1. \quad (5)$$

This is the standard Bloch/Fano two-qubit decomposition with the one-body Bloch vectors fixed by the LT constraint [9].

Restricting to  $T = \text{diag}(t_x, t_y, t_z)$  gives an  $X$ -state, standard in the two-qubit literature

[10, 11], and in the LT setting that restriction produces

$$\rho = \begin{pmatrix} g^2 + \frac{t_z}{4} & 0 & 0 & \frac{t_x - t_y}{4} \\ 0 & g(1 - g) - \frac{t_z}{4} & \frac{t_x + t_y}{4} & 0 \\ 0 & \frac{t_x + t_y}{4} & g(1 - g) - \frac{t_z}{4} & 0 \\ \frac{t_x - t_y}{4} & 0 & 0 & (1 - g)^2 + \frac{t_z}{4} \end{pmatrix}. \quad (6)$$

Equation (6) is therefore not quoted as a black-box formula. It follows directly by inserting the LT Bloch vectors into (5) and then taking diagonal  $T$ .

**Proposition 3.3** (Diagonal- $T$  feasibility and separability). *The diagonal- $T$  LT slice is feasible exactly when*

$$\begin{aligned} g^2 + \frac{t_z}{4} &\geq 0, & (1 - g)^2 + \frac{t_z}{4} &\geq 0, \\ (4g^2 + t_z)(4(1 - g)^2 + t_z) &\geq (t_x - t_y)^2, & t_z + |t_x + t_y| &\leq 4g(1 - g). \end{aligned}$$

Because PPT is equivalent to separability for  $2 \times 2$  systems [12, 13], the separable part of the slice is obtained by adjoining the companion conditions

$$(4g^2 + t_z)(4(1 - g)^2 + t_z) \geq (t_x + t_y)^2, \quad t_z + |t_x - t_y| \leq 4g(1 - g).$$

This slice was therefore used as a solvable warmup family and as a clean benchmark for the SDP code. It captures classical LT, separable LT, and entangled LT inside one exactly computable three-parameter family.

**Infinite-temperature limit.** At  $\beta = 0$  one has  $g = \frac{1}{2}$  and  $\gamma = I/2$ . Then the same inequalities reduce to

$$1 - t_z \geq |t_x + t_y|, \quad 1 + t_z \geq |t_x - t_y|,$$

which are equivalent to the Bell-diagonal tetrahedron. The PPT/separability inequalities reduce further to

$$|t_x| + |t_y| + |t_z| \leq 1,$$

which is the Bell-diagonal octahedron. This limiting case is useful because it provides a familiar reference geometry against which the finite-temperature LT slice can be checked.

**LT rays.** For any zero-marginal Hermitian direction  $C_0$ , the family

$$\rho(p) = \Gamma + pC_0$$

remains in the affine LT space. Whitening by  $\Gamma^{-1/2}$  gives

$$\rho(p) \succeq 0 \iff I + p \Gamma^{-1/2} C_0 \Gamma^{-1/2} \succeq 0,$$

so the exact feasible interval for  $p$  is obtained from the eigenvalues of the whitened direction. This is why rays were useful numerically: they give exact one-parameter LT families with known boundaries.

### 3.5 Qutrits: full LT body versus structured subclass

The full symmetric qutrit LT set has dimension 64, so a qubit-style full classification is not realistic within one capstone. Still, the same general framework applies:

$$\text{LT}(\gamma, \gamma) = (\Gamma + \mathcal{Z}) \cap \text{PSD}, \quad \dim_{\mathbb{R}} \text{LT} = 64.$$

So the full qutrit LT body is again a spectrahedron, just in a much larger affine space. This is why the qutrit part of the project shifts from full classification to structured subclasses.

Let

$$\gamma = \text{diag}(g_1, g_2, g_3), \quad g_i > 0, \quad \sum_i g_i = 1.$$

A commuting qutrit LT state is equivalent to a nonnegative matrix  $P = (p_{ij})$  with row and column sums both equal to  $g$ :

$$\sum_j p_{ij} = g_i, \quad \sum_i p_{ij} = g_j.$$

This is exactly the classical transportation-polytope setup discussed by De Loera and Kim [16]. Solving the linear row/column-sum constraints explicitly gives the report's parameterisation

$$P = \begin{pmatrix} a & b & g_1 - a - b \\ c & d & g_2 - c - d \\ g_1 - a - c & g_2 - b - d & g_3 - g_1 + a + b + c + d \end{pmatrix}. \quad (7)$$

So again the displayed formula is not an unexplained ansatz: it is the standard  $3 \times 3$  transportation-polytope parameterisation with Gibbs margins.

Its dimension is therefore 4. For this commuting subclass, mutual information reduces to the classical quantity

$$I(P) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{g_i g_j} = 2H(g) - H(P).$$

Thus

$$0 \leq I(P) \leq H(g),$$

with minimum at the product coupling  $P = gg^T$  and maximum at permutation couplings. This gives a nontrivial exact qutrit result without claiming a full 64-dimensional classification.

A useful extra structural fact is that the faces of the transportation polytope are determined by support patterns. If  $S \subseteq \{1, 2, 3\} \times \{1, 2, 3\}$  is the support graph, then the corresponding face dimension is

$$\dim F_S = |S| - 6 + c(S),$$

where  $c(S)$  is the number of connected components of the support graph. This explains why the commuting qutrit LT family has a combinatorial boundary structure: unlike the full qutrit spectrahedron, its faces are controlled by which matrix entries are forced to zero.

### 3.6 Direct source map for the three main displayed parameterisations

Because these formulas are exactly the kind of thing one gets asked about in a meeting, it is worth stating explicitly where they come from.

First, the Pauli-correlation formula

$$\rho = \frac{1}{4} \left( I \otimes I + z_\gamma (\sigma_z \otimes I + I \otimes \sigma_z) + \sum_{i,j} T_{ij} \sigma_i \otimes \sigma_j \right)$$

is the LT-specialised form of the standard Bloch/Fano decomposition of a two-qubit state [9]. The only project-specific step is the substitution of the thermal marginals, which forces the local Bloch vectors to be  $(0, 0, z_\gamma)$ .

Second, the  $X$ -state matrix

$$\rho = \begin{pmatrix} g^2 + \frac{t_z}{4} & 0 & 0 & \frac{t_x - t_y}{4} \\ 0 & g(1 - g) - \frac{t_z}{4} & \frac{t_x + t_y}{4} & 0 \\ 0 & \frac{t_x + t_y}{4} & g(1 - g) - \frac{t_z}{4} & 0 \\ \frac{t_x - t_y}{4} & 0 & 0 & (1 - g)^2 + \frac{t_z}{4} \end{pmatrix}$$

comes from imposing diagonal  $T$  inside that same Bloch/Fano representation. The general two-qubit  $X$ -state class is standard in the literature [10, 11]; the project-specific step is the LT specialisation with thermal local bias.

Third, the commuting-qutrit matrix

$$P = \begin{pmatrix} a & b & g_1 - a - b \\ c & d & g_2 - c - d \\ g_1 - a - c & g_2 - b - d & g_3 - g_1 + a + b + c + d \end{pmatrix}$$

is simply the explicit solution of the  $3 \times 3$  row-sum and column-sum constraints for a transportation polytope with margins  $g$  [16]. So the underlying idea is classical polytope theory; the thermodynamic content comes from interpreting the margins as Gibbs probabilities.

## 4 Operational results and SDP methods

### 4.1 An explicit ordered LT family

Let

$$|\tilde{\gamma}\rangle = \sum_i \sqrt{g_i} |i\rangle \otimes |i\rangle, \quad \tilde{\gamma} := |\tilde{\gamma}\rangle\langle\tilde{\gamma}|,$$

and let  $\Delta$  denote full product-energy dephasing. Then

$$\tilde{\gamma}_{\text{diag}} := \Delta(\tilde{\gamma}) = \sum_i g_i |ii\rangle\langle ii|.$$

Both states are LT. Define the interpolation family

$$\rho_\lambda = (1 - \lambda)\tilde{\gamma} + \lambda\tilde{\gamma}_{\text{diag}}, \quad 0 \leq \lambda \leq 1.$$

Now set

$$G_\mu = (1 - \mu)\text{id} + \mu\Delta, \quad 0 \leq \mu \leq 1.$$

Because dephasing is local GP, so is  $G_\mu$ , and direct substitution gives

$$G_\mu(\rho_\lambda) = \rho_{\lambda'}, \quad \lambda' = \lambda + \mu(1 - \lambda).$$

Hence whenever  $0 \leq \lambda \leq \lambda' \leq 1$ ,

$$\rho_\lambda \succ_{\text{LGP}} \rho_{\lambda'}.$$

This family is the clean analytic example showing that local GP can decrease LT correlations in a fully explicit way.

In the product energy basis the map is completely transparent:  $\Delta$  removes the off-diagonal coherence terms while leaving the diagonal populations and hence the local Gibbs marginals unchanged. That is exactly why this family is useful in the report: it gives a constructive

local GP transformation, not just an existence statement.

## 4.2 Why SDPs are the right numerical tool

The geometry of LT is “PSD plus linear equalities”, and GP convertibility in the Choi picture is also “PSD plus linear equalities”. That is exactly semidefinite programming.

**LT membership.** Given a candidate state  $\rho$ , LT membership is the feasibility problem

$$\rho \succeq 0, \quad \text{Tr}_A \rho = \gamma, \quad \text{Tr}_B \rho = \gamma, \quad \text{Tr} \rho = 1.$$

**Projection to LT.** To project a sampled state toward LT, one solves a convex optimisation such as

$$\min_{\sigma \in \text{LT}(\gamma)} \frac{1}{2} \|\rho - \sigma\|_1.$$

This is used to generate clean LT test states before convertibility analysis.

**Global GP convertibility in Choi form.** The standard representation of a channel by a Choi matrix goes back to Choi [14]; semidefinite formulations built from this representation are standard in quantum information [15]. A channel  $\Phi$  from input space to output space is represented by its Choi matrix  $J_\Phi$ . Global GP convertibility from  $\rho$  to  $\sigma$  asks for

$$J_\Phi \succeq 0, \quad \text{Tr}_{\text{out}} J_\Phi = I_{\text{in}},$$

plus the linear constraints encoding

$$\Phi(\rho) = \sigma, \quad \Phi(\Gamma) = \Gamma.$$

Using the standard Choi action formula,

$$\Phi(X) = \text{Tr}_{\text{in}} \left[ J_\Phi (I_{\text{out}} \otimes X^T) \right],$$

these become affine constraints in  $J_\Phi$ , so the whole problem is an SDP.

In practice this means the solver never searches over Kraus operators directly. It searches over a positive semidefinite matrix subject to linear equalities. That is the exact reason the global convertibility problem is numerically stable enough to be a central computational tool in the project.

**Local GP convertibility and why relaxations are needed.** Here one asks for channels  $\Phi_A$  and  $\Phi_B$  such that

$$\Phi_A(\gamma) = \gamma, \quad \Phi_B(\gamma) = \gamma, \quad (\Phi_A \otimes \Phi_B)(\rho) = \sigma.$$

The difficulty is the tensor-product restriction on the channel. In Choi variables this becomes a product constraint,

$$J_{AB} = J_{\Phi_A} \otimes J_{\Phi_B},$$

which is nonconvex. That is the precise mathematical reason local convertibility is harder than global convertibility: the feasible set is no longer described by one PSD cone plus linear equations.

The code therefore uses two levels:

- (a) a *verified local* test for report-facing local edges, and
- (b) a PPT-type outer relaxation as a convex upper bound on the local feasible set.

The reason for using a relaxation is structural, not ad hoc. The set of product Choi matrices is not convex, but a PPT-style enlargement is still semidefinite and therefore computationally tractable. So the relaxation is used to bound local convertibility from above, while final local claims in the report are kept for verified constructions.

### 4.3 Numerical conventions used in the code

The reported numerics use LT benchmark states produced either from analytic families or by convex projection onto LT. Global convertibility is solved directly as a Choi SDP. Local report-facing claims are based on verified local constructions rather than on the relaxation alone. The PPT relaxation is therefore interpreted as an outer bound: a positive PPT edge means “not ruled out by the relaxation”, not “proved local”. This distinction is essential when reading the convertibility plots.

## 5 Core numerical results

Only the main report-facing figures are kept in the body; secondary diagnostics are moved to the appendix.

## 5.1 Geometry of sampled LT states

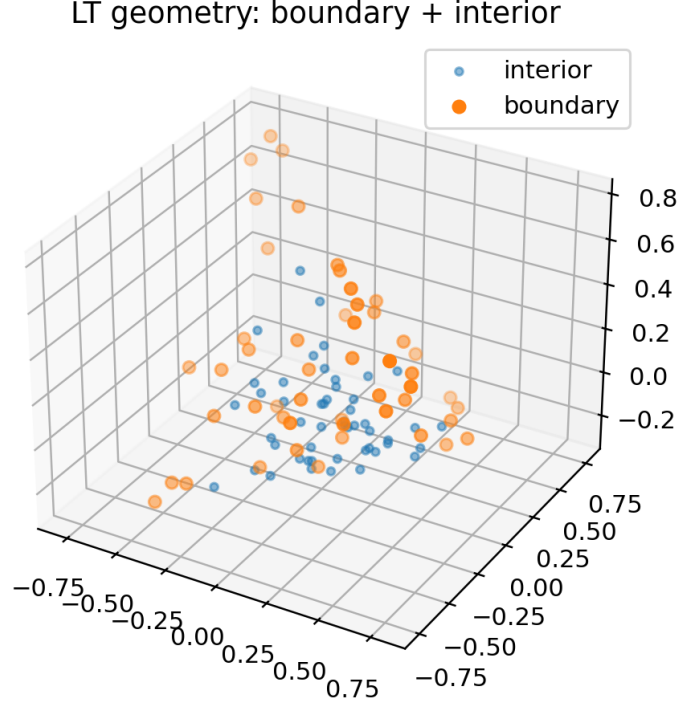


Figure 1: Sampled LT geometry in a low-dimensional embedding. The figure is used qualitatively: the product Gibbs state sits near the low-correlation centre, while boundary points correspond to more strongly constrained or more strongly correlated LT states.

This figure is not a proof of global geometry by itself. Its role is to show that the LT set is a nontrivial constrained body rather than a one-parameter family, and to support the distinction between interior LT states and boundary states where positivity becomes tight.

## 5.2 Correlation equals athermality along LT thermalisation paths

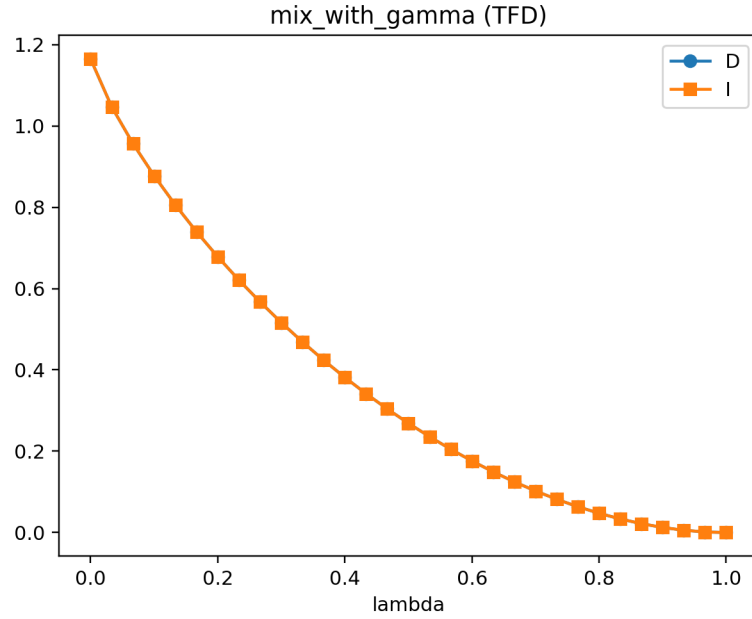


Figure 2: Mixing path from the TFD state to the product Gibbs state. Along this LT path, the relative entropy to equilibrium and the mutual information coincide numerically throughout.

Along the LT path from TFD to  $\Gamma$ , the curves for  $D(\rho||\Gamma)$  and  $I(A:A')$  lie on top of one another and decrease to zero at equilibrium. This is the direct numerical confirmation of the analytic identity  $D = I$  on LT.

### 5.3 PPT classification inside sampled LT

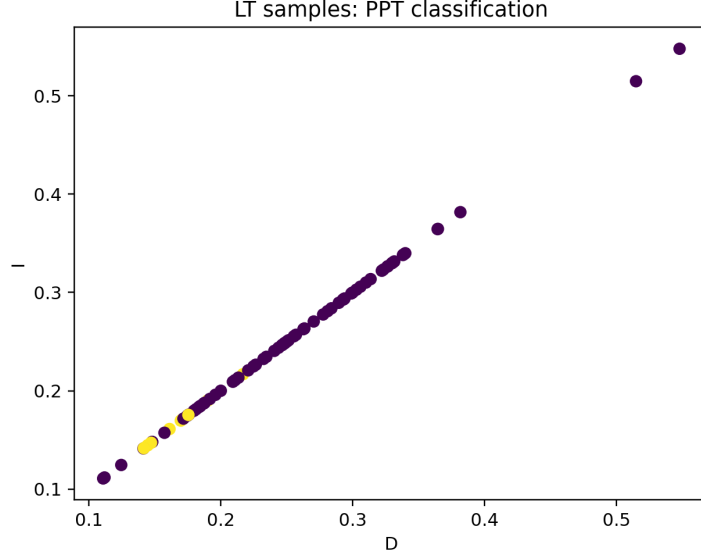


Figure 3: PPT classification of sampled two-qubit LT states in the  $(D, I)$  plane. Since all samples are LT, the points lie on the diagonal  $D = I$ .

This figure is useful for separating two distinct statements. The equality  $D = I$  is an LT identity and therefore holds for every plotted state, regardless of whether the state is separable or entangled. The PPT label gives extra structural information inside LT: low-correlation states are often PPT in this sampling, while stronger correlations more frequently reach the non-PPT region.

### 5.4 Global versus local convertibility

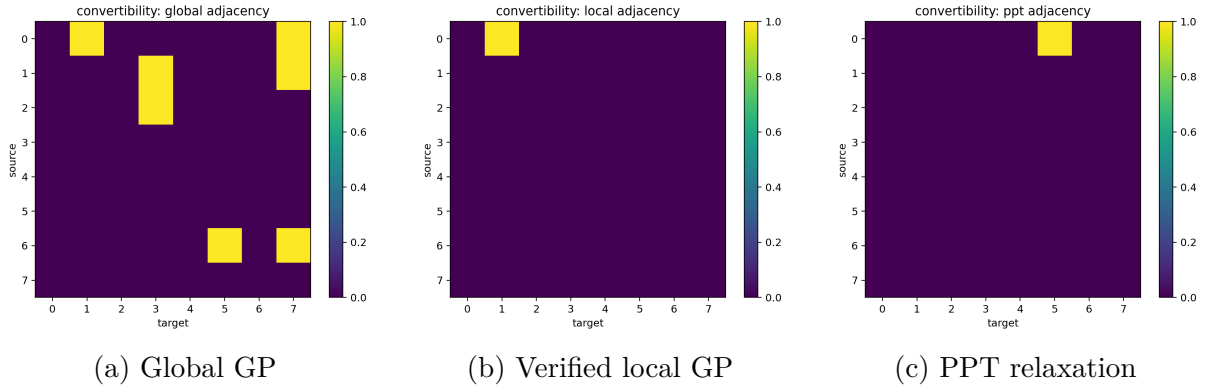


Figure 4: Benchmark convertibility comparison. Global GP produces a much denser adjacency pattern than verified local GP, while the PPT relaxation gives an intermediate outer approximation.

This is the main operational figure. Its message is not that PPT and verified local coincide; they do not. The message is that global GP admits many more edges than local GP, while the PPT relaxation sits in between as an efficiently computable outer bound. That is exactly the hierarchy predicted by the channel geometry discussed above.

## 5.5 How to read the adjacency plots

Each convertibility heatmap is an adjacency matrix for an ordered family of benchmark LT states. The row index is the source state and the column index is the target state. A marked off-diagonal entry means that the corresponding pipeline found a channel taking the row state to the column state. The diagonal is the trivial identity map and therefore carries no substantive thermodynamic information; the relevant content is the off-diagonal pattern.

This viewpoint is useful because it converts a long list of pairwise feasibility results into a single geometric object. Dense off-diagonal structure means many ordered pairs are comparable under the chosen class of operations. Sparse off-diagonal structure means most pairs are incomparable. The global panel is visibly denser because its feasible set is the full convex family of GP channels. The verified-local panel is much sparser because the product-channel constraint removes most of those feasible directions.

The PPT panel must be read differently. It is not a proof of local convertibility. It is an efficiently computable outer test that keeps the SDP convex. Therefore the correct interpretation is one-sided: an edge absent already at the PPT level is strongly ruled out by the relaxation, whereas an edge present only in the PPT panel is merely compatible with the relaxation and still requires verified-local confirmation before it can be promoted to a local claim.

## 6 Discussion

### 6.1 What has been achieved

The project delivers three connected results. First, LT has been identified as a structured convex state space rather than a loose collection of examples: it is a spectrahedron with a precise dimension formula and a natural correlation-tensor parameterisation. Second, on LT the free-energy excess above  $\Gamma$  is exactly the mutual information, so the remaining thermodynamic resource has a direct information-theoretic interpretation. Third, the convertibility problem splits sharply between global and local GP maps: global GP is convex in the Choi picture, whereas local GP becomes nonconvex because of the product-channel restriction.

For qubits, the report goes beyond warmup slices. The full symmetric two-qubit LT problem has been reduced to a 9-parameter normal form, and generic positivity has been converted into explicit Schur-complement inequalities. This is the main structural theorem because it replaces an abstract positivity problem on  $4 \times 4$  density matrices by concrete inequalities in the independent LT parameters.

The slice and ray analyses still matter because they are the route by which the full body was first explored. The diagonal- $T$  family gives an exactly solvable three-parameter benchmark with explicit positivity and separability tests, while LT rays give one-parameter families with exact endpoints determined by the whitened direction. These subfamilies were useful both analytically and numerically before the full qubit description was developed.

### 6.2 How the three objectives were met

**Objective 1: LT structure.** The LT set has been identified as a spectrahedron, its exact dimension has been derived, and the full symmetric two-qubit problem has been reduced to an explicit 9-parameter family with a generic Schur-complement positivity test. The commuting qutrit subclass adds a higher-dimensional structured example with exact polytope geometry.

**Objective 2: thermodynamic meaning.** On LT, the relative entropy to the product Gibbs state is exactly mutual information. This turns correlation into the entire nonequilibrium resource once local marginals are fixed to equilibrium.

**Objective 3: convertibility.** Global GP convertibility is naturally convex in the Choi picture, while local GP introduces a nonconvex product-channel restriction. Both analytically and numerically, the local theory is far more restrictive. The TFD-to-dephased

family gives a clean positive local example; the benchmark adjacency plots show that most global edges disappear once locality is imposed.

### 6.3 Limitations and next steps

The main limitation is dimensional growth. The qubit problem is still explicit; the full qutrit problem is already too large for a comparable analytic classification, which is why the commuting qutrit subclass is the appropriate higher-dimensional contribution. The next natural step is to sharpen necessary conditions for local convertibility and to expand the verified-local numerical study around the analytic families already identified.

## 7 Conclusion

This project treats locally thermal bipartite states as a setting in which all local athermality has been removed and any remaining thermodynamic resource is necessarily global. The resulting LT body is convex, constrained, and operationally meaningful. On this manifold, relative entropy to product thermal equilibrium is exactly mutual information, so correlations acquire a direct thermodynamic interpretation.

The principal analytic outcome is the near-complete treatment of the two-qubit problem. The full symmetric two-qubit LT set is a 9-dimensional spectrahedron, and its generic positivity can be reduced to explicit Schur-complement inequalities. Supporting this, the diagonal- $T$  family and ray constructions supply exact warmup analyses, including positivity, separability, and monotonicity tests. In higher dimension, the commuting qutrit LT subclass gives a tractable and nontrivial analytic extension with exact polytope structure and exact mutual-information range.

On the operational side, the report distinguishes global and local Gibbs-preserving convertibility, proves that LT is closed under local GP maps, and exhibits an explicit totally ordered TFD-to-dephased chain under local GP. Together with the SDPs implemented in the accompanying code, this provides a coherent framework for analysing both geometry and dynamics of locally thermal states.

## A Appendix: detailed warmup-family formulas

### A.1 Diagonal- $T$ feasibility and separability

For two qubits, restricting the Pauli tensor to

$$T = \text{diag}(t_x, t_y, t_z)$$

gives the  $X$ -state

$$\rho = \begin{pmatrix} g^2 + \frac{t_z}{4} & 0 & 0 & \frac{t_x - t_y}{4} \\ 0 & g(1 - g) - \frac{t_z}{4} & \frac{t_x + t_y}{4} & 0 \\ 0 & \frac{t_x + t_y}{4} & g(1 - g) - \frac{t_z}{4} & 0 \\ \frac{t_x - t_y}{4} & 0 & 0 & (1 - g)^2 + \frac{t_z}{4} \end{pmatrix}.$$

Positivity is equivalent to positivity of the two  $2 \times 2$  diagonal blocks, which yields

$$\begin{aligned} g^2 + \frac{t_z}{4} &\geq 0, & (1 - g)^2 + \frac{t_z}{4} &\geq 0, \\ (4g^2 + t_z)(4(1 - g)^2 + t_z) &\geq (t_x - t_y)^2, & t_z + |t_x + t_y| &\leq 4g(1 - g). \end{aligned}$$

Because PPT is equivalent to separability in  $2 \times 2$ , the partial transpose gives the companion inequalities

$$(4g^2 + t_z)(4(1 - g)^2 + t_z) \geq (t_x + t_y)^2, \quad t_z + |t_x - t_y| \leq 4g(1 - g),$$

which together give the exact separable subset of the diagonal- $T$  LT slice.

### A.2 Exact LT ray bounds

For any zero-marginal Hermitian direction  $C_0$ , consider

$$\rho(p) = \Gamma + pC_0.$$

Then

$$\rho(p) \succeq 0 \iff I + p\tilde{C}_0 \succeq 0, \quad \tilde{C}_0 := \Gamma^{-1/2}C_0\Gamma^{-1/2}.$$

If  $\{\lambda_k\}$  are the eigenvalues of  $\tilde{C}_0$ , the exact feasible interval is

$$p \in \left[ \max_{\lambda_k > 0} \left( -\frac{1}{\lambda_k} \right), \min_{\lambda_k < 0} \left( -\frac{1}{\lambda_k} \right) \right].$$

This is why rays are analytically clean and numerically stable.

## B Appendix: qutrit combinatorics kept out of the main body

For the commuting qutrit transportation polytope  $\mathcal{T}(g, g)$ , a support pattern  $S \subseteq \{1, 2, 3\} \times \{1, 2, 3\}$  determines a face whose dimension is

$$\dim F_S = |S| - 6 + c(S),$$

where  $c(S)$  is the number of connected components of the support graph. Vertices are exactly feasible couplings with acyclic support graphs. These details are mathematically useful but not needed in the main narrative once the polytope interpretation and the mutual-information range have been established.

## C Appendix: additional numerical figures

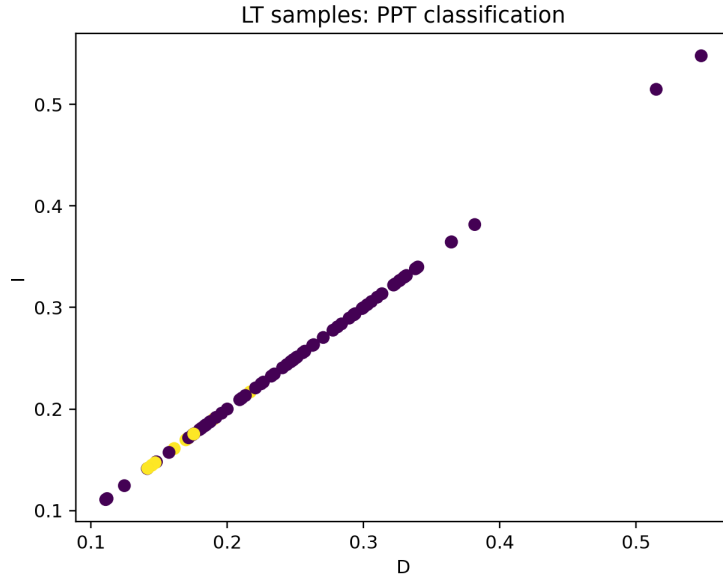


Figure 5: PPT classification of sampled two-qubit LT states in the  $(D, I)$  plane. Since all samples are LT, the points lie on the diagonal  $D = I$ . PPT states cluster near the low-correlation region for this sampling rule.

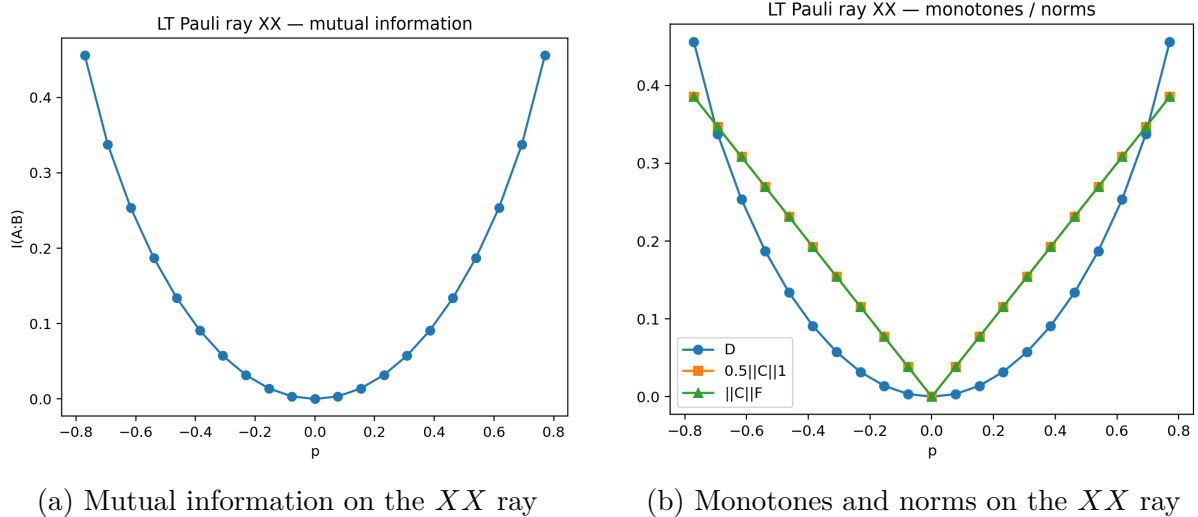


Figure 6: Resource profile along the Pauli-XX LT ray. The minimum occurs at the product Gibbs point and the quantities grow with  $|p|$ .

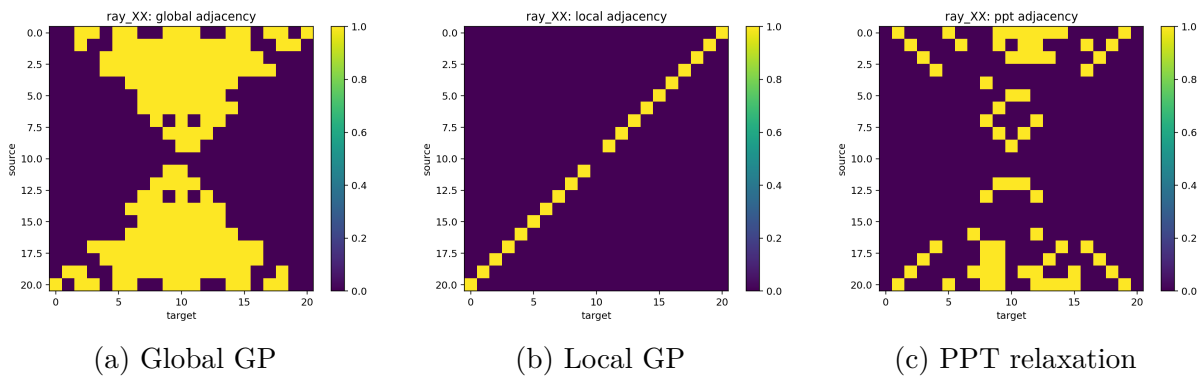


Figure 7: Convertibility structure on the Pauli-XX LT ray. Global GP shows many inward-directed edges, while verified local GP is much more restricted and the PPT relaxation sits in between.

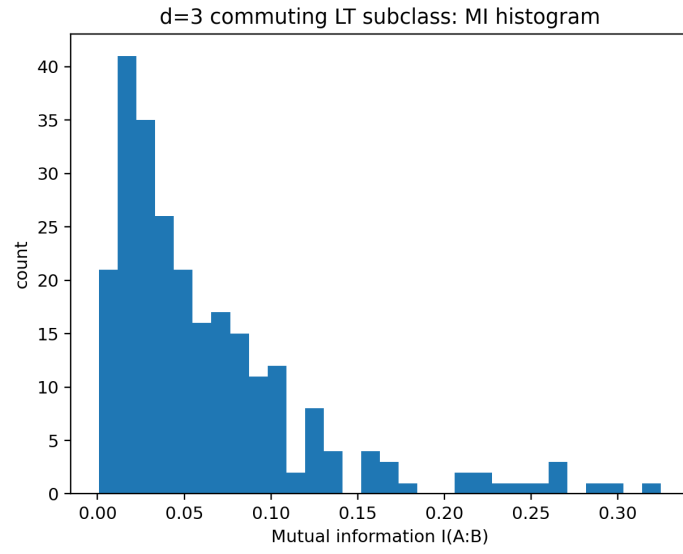


Figure 8: Mutual-information histogram for sampled states in the commuting qutrit LT subclass.

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